Abdul Muneem

Ph.D. in Engineering Sciences (Applied Physics)

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Education:

Ph.D. Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (January-2018 to December-2023), Pakistan, and High Energy Nuclear Physics Laboratory, RIKEN, Japan as International Program Associate.

Thesis title: Development of precise neutron imaging with nuclear emulsion and fluorescent nuclear track detectors

Ph.D. Supervisor: Prof. Jameel-Un Nabi, VC, University of Wah, Pakistan

Ph.D. Co-supervisor: Prof. Takehiko R. Saito, Chief Scientist, RIKEN, Japan

MS (Physics) COMSATS Institute of Information Technology, Pakistan (August-2014 to June-2016).

Thesis title: Electromagnetically Induced Chirality in a Four Levels Atomic Medium and the Retrieved Light Pulse

BS (Physics) University of Peshawar, Pakistan (September-2009 to January-2014).

Thesis title: Wind Power Potential in Pakistan

Positions and Employment:

August 2024 – to date Postdoctoral Researcher, Neutron Beam Technology Team, Center for Advanced Photonics, RIKEN, Japan.

Scholarships:

i. International Program Associate, IPA, at High Energy Nuclear Physics Laboratory, RIKEN, Japan (July-2019 to December-2022).
ii. Higher Education Commission, Pakistan, Pak-Turk Researchers Mobility Grant (Phase-I), Karabuk University, Turkey (August-2018 to December-2018).
iii. Graduate Assistantship at GIK Institute, Pakistan (January-2018 to December-2023).
iv. Research Assistant in Higher Education Commission, Pakistan (NRPU) project at GIK Institute, Pakistan (January-2018 to December-2019).

Professional Memberships:

i. Japan Physical Society (May-2020 – December-2022).
ii. Japanese Society for Neutron Science (October-2024 – to date)

Talks and Posters: Conference, Symposium, Workshop, School:

i. 9th RIKEN-JCNS Joint Workshop 2025, Riken, Japan (16-18 June 2025), **Title**: Recent Progress in Developing High-Resolution Neutron Imaging Using FGNEs and FNTDs.

ii. 11th International Meeting of the Union for Compact Accelerator-driven Neutron Sources, TRIUMF, Canada's Particle Accelerator Centre, Vancouver, Canada (24-28 February 2025). **Title**: Advancement of Neutron Imaging Techniques Towards the Highest Resolution and Development of Imaging Application with RANS Using Particle Tracking Detectors.

iii. The 8th Neutron and Muon School, J-PARC MLF & JRR-3, Tokai, Ibaraki, Japan (09-13 December 2024). (**Attend**)

iv. Japan Neutron Science Society 24th Annual Meeting, Nagoya International Congress Center, Nagoya, Japan (04-06 December 2024). **Title**: Advancing Neutron Imaging Techniques to Highest Resolution Using Particle Tracking Detectors.

v. RANS symposium, RIKEN, Japan (19 November 2024). **Poster title**: Advancing Neutron Imaging Techniques to Highest Resolution with Solid-State Track Detectors.

vi. The 6th workshop on quantum beam imaging (QBI2024), Tokyo university of science, Japan (26-27 September 2024). **Title**: Advancing Neutron Imaging Techniques to Highest Resolution with Solid-State Track Detectors.

vii. 8th Multi-Disciplinary Student Research International Conference MDSRIC – 2023, University of Wah, Pakistan (6-7 December 2023). **Title**: Recent developments of neutron imaging techniques using solid-state track detectors.

viii. Research talks at Faculty of Engineering Sciences, GIK institute, Pakistan (17 October 2023). **Title**: Recent developments of neutron imaging techniques using nuclear emulsion.

ix. 35th Workshop on Solid State Nuclear Track Detectors, Kobe university, Japan (28-29 March 2023). **Title**: Recent developments of neutron imaging techniques using solid-state track detectors.

x. KEK S-type workshop, Japan (15-16 December 2022). **Title**: Developing neutron imaging techniques using nuclear emulsion and fluorescent nuclear track detectors.

xi. The RIKEN-CPR Physics Workshop for Students, Japan (12-13 December 2022). **Title**: Recent developments of neutron imaging techniques using solid-state track detectors.

xii. The 4th workshop on quantum beam imaging (QBI2022), Riken, Japan (26-27 September 2022). **Title**: Development of neutron imaging techniques using nuclear emulsion and fluorescent nuclear track detectors.

xiii. International Conference on Material and systems for sustainability (ICMaSS2021), Japan (4-6 November 2021). **Title**: Development of precise neutron imaging technique using fine grained Nuclear Emulsion and Machine Learning.

xiv. JPS 2021 annual meeting (Autumn), Japan (14-17 September 2021). **Title**: Progress in precise Neutron imaging using Nuclear Emulsion and Machine Learning.

xv. JPS 2020 annual meeting (Autumn), Japan (14-17 September 2020). **Title**: Image processing for cold/ultracold neutron detection with fine-grained nuclear emulsion.

xvi. RIKEN Summer School, Kisarazu, Chiba, Japan (7-8 October 2019). **Poster title**: Development of Scintillating Fiber Detectors for the hypernuclear experiment with the fragment separator FRS at GSI.

xvii. Sixth Italian Pakistani workshop on Relativistic Astrophysics, NUST, Pakistan (24-26 January 2019). **Title**: Gamow-Teller and first forbidden transitions for the r-process N=82 waiting point nuclei.

xviii. 5th International Symposium on Multidisciplinary Studies, Ankara, Turkey (16-17 November 2018). **Title**: Beta Decay Half-lives of some Radioisotopes Produced in a Thermonuclear Reactor.

xix. International Conference on Nuclear Structure Properties (NSP2018), Karadeniz Technical University, Trabzon, Turkey (12-14 September 2018). **Poster title**: First Forbidden β-Decay Logft Values of Neutron Rich Te Isotopes.

Publications and Patents:

i. A. Kasagi, T. R. Saito, V. Drozd, H. Ekawa, S. Escrig, Y. Gao, Y. He, E. Liu, **A. Muneem**, M. Nakagawa, K. Nakazawa, C. Rappold, N. Saito, M. Taki, Y. K. Tanaka, H. Wang, A. Yanai, J. Yoshida, and M. Yoshimoto, Binding energy of ${}_{\Lambda}^{3}$H and ${}_{\Lambda}^{4}$H via image analyses of nuclear emulsions using deep-learning, Progress of Theoretical and Experimental Physics (PTEP), 2025, https://doi.org/10.1093/ptep/ptaf097

ii. Yan He, Vasyl Drozd, Hiroyuki Ekawa, Samuel Escrig, Yiming Gao, Ayumi Kasagi, Enqiang Liu, **Abdul Muneem**, Manami Nakagawa , Kazuma Nakazawa, Christophe Rappold, Nami Saito, Takehiko R. Saito, Shohei Sugimoto, Masato Taki , Yoshiki K. Tanaka, He Wang , Ayari Yanai, Junya Yoshida, Hongfei Zhanga, A novel machine learning method to detect double-Λ hypernuclear events in nuclear emulsions, Nuclear Instruments and Methods in Physics Research, Section A, 1073 (2025) 170196, https://doi.org/10.1016/j.nima.2024.170196

iii. **Abdul Muneem**, Junya Yoshida, Takehiko R. Saito, Hiroyuki Ekawa, Masahiro Hino, Katsuya Hirota, Go Ichikawa, Ayumi Kasagi, Masaaki Kitaguchi, Kenji Mishima, Jameel-Un Nabi, and Manami Nakagawa, Advancing Neutron Imaging Techniques to Highest Resolution with Fluorescent Nuclear Track Detectors, Scientific Reports, 15, 2103 (2025), https://doi.org/10.1038/s41598-024-84591-x

iv. **Patent**: Title of invention: 中性子位置記録装置、飛跡画像生成装置、及び飛跡画像生成方法, Date of submission: 25th December 2024.

v. Abdul Kabir, Jameel-Un Nabi, Muhammad Tahir, **Abdul Muneem** and Zain Ul Abideen, Re-analysis of Temperature Dependent Neutron Capture Rates and Stellar β−-decay Rates of ${}^{95-98}$Mo, Chinese Physics C, 2024, 48(9): 094101, https://doi.org/10.1088/1674-1137/ad5428

vi. E. Liu, V. Drozd, H. Ekawa, S. Escrig, Y. Gao, Y. He, A. Kasagi, M. Nakagawa, H. Ong, C. Rappold, R. Sekiya, T. R. Saito, X. Tang, Y. K. Tanaka, H. Wang, A. Yanai, P. Achenbach, H. Alibrahim Alfaki, F. Amjad, M. Armstrong, K.-H. Behr, J. Benlliure, Z. Brencic, T. Dickel, S. Dubey, M. Feijoo-Fontán, H. Fujioka, H. Geissel, F. Goldenbaum, A. Graña Gonzáler, E. Haettner, M. N. Harakeh, H. Heggen, C. Hornung, N. Hubbard, K. Itahashi, M. Iwasaki, N. Kalantar-Nayestanaki, M. Kavatsyuk, E. Kazantseva, A. Khreptak, B. Kindler, R. Knoebel, H. Kollmus, D. Kostyleva, S. Kraft-Bermuth, N. Kurz, B. Lommel, S. Minami, D. J. Morrissey, P. Moskal, I. Mukha, **A. Muneem**, K. Nakazawa, C. Nociforo, S. Pietri, J. Pochodzalla, S. Purushothaman, E.

Rocco, J. L. Rodríguez-Sánchez, P. Roy, R. Ruber, S. Schadmand, C. Scheidenberger, P. Schwarz, V. Serdyuk, M. Skurzok, B. Streicher, K. Suzuki, B. Szczepanczyk, N. Tortorelli, M. Vencelj, T. Weber, H. Weick, M. Will, K. Wimmer, A. Yamamoto,J. Yoshida and J. Zhao, A compact start time counter using plastic scintillators readout with MPPC arrays for the WASA-FRS HypHI experiment, Nuclear Instruments and Methods in Physics Research, Section A, 1064 (2024) 169384, https://doi.org/10.1016/j.nima.2024.169384

vii. A. Kasagi, W. Dou, V. Drozd, H. Ekawa, S. Escrig, Y. Gao, Y. He, E. Liu, **A. Muneem**, M. Nakagawa, K. Nakazawa, C. Rappold, N. Saito, T.R. Saito, S. Sugimoto, M. Taki, Y.K. Tanaka, A. Yanai, J. Yoshida M. Yoshimoto, and H. Wang, Hypernuclear event detection in the nuclear emulsion with Monte Carlo simulation and machine learning, Nuclear Instruments and Methods in Physics Research, Section A, 1056 (2023) 168663, https://doi.org/10.1016/j.nima.2023.168663

viii. T. R. Saito, *et al,* The WASA-FRS project at GSI and its perspective, Nuclear Inst. and Methods in Physics Research, B 542 (2023) 22–25, https://doi.org/10.1016/j.nimb.2023.05.042

ix. Hiroyuki Ekawa, Wenbou Dou, Yiming Gao, Yan he, Aymikasagi, Enqiang Liu, **Abdul Muneem,** Manami Nakagawa, Christophe Rappold, Nami Saito, Takehiko R. Saito, Masato Taki, Yoshiki K. Tanaka, He Wang, Junya Yoshida, Development of machine learning analyses with graph neural network for the WASA-FRS experiment, European Physical Journal A, 59, 103,(2023), https://dx.doi.org/10.1140/epja/s10050-023-01016-5

x. **Abdul Muneem**, Junya Yoshida, Hiroyuki Ekawa, Masahiro Hino, Katsuya Hirota, Go Ichikawa, Ayumi Kasagi, Masaaki Kitaguchi, Naoto Muto, Kenji Mishima, Jameel-Un Nabi, Manami Nakagawa, Naotaka Naganawa, and Takehiko R. Saito, Investigation of neutron imaging applications using fine-grained nuclear emulsion, Journal of Applied Physics, 133, 054902 (2023), https://doi.org/10.1063/5.0131098

xi. **Abdul Muneem**, Junya Yoshida, Hiroyuki Ekawa, Masahiro Hino, Katsuya Hirota, Go Ichikawa, Ayumi Kasagi, Masaaki Kitaguchi, Satoshi Kodaira, Kenji Mishima, Jameel-Un Nabi, Manami Nakagawa, Michio Sakashita, Norihito Saito, Takehiko R. Saito, Satoshi Wada, Nakahiro Yasuda, Study on the reusability of fluorescent nuclear track detectors using optical bleaching, Radiation Measurements, 158, 106863, (2022), https://doi.org/10.1016/j.radmeas.2022.106863

xii. Jameel-Un Nabi, Asim Ullah, Abdul Kabir, **Abdul Muneem** and Mahmut Böyükata, Investigation of Gamow-Teller strength of ^{186}Hg within deformed pn-QRPA, Canadian Journal of Physics, 100(7): 329-333. (2022) https://doi.org/10.1139/cjp-2021-0415

xiii. Takehiko R. Saito, Wenbou Dou, Vasyl Drozd, Hiroyuki Ekawa , Samuel Escrig , Yan He, Nasser Kalantar-Nayestanaki, Ayumi Kasagi, Myroslav Kavatsyuk, Enqiang Liu, Yue Ma, Shizu Minami, **Abdul Muneem**, Manami Nakagawa, Kazuma Nakazawa, Christophe Rappold, Nami Saito, Christoph Scheidenberger, Masato Taki, Yoshiki K. Tanaka, Junya Yoshida, Masahiro Yoshimoto, He Wang and Xiaohong Zhou, New directions in hypernuclear physics, Nature Reviews Physics, 3, 803-813 (2021), https://doi.org/10.1038/s42254-021-00371-w

Conference proceedings:

i. Abdul Kabir, **Abdul Muneem** and Jameel-Un Nabi, Re-examining Neutron Capture and Stellar Weak Rates of ^{97}Mo under Astrophysical Conditions, Cumhuriyet Sci. J., 46(3) (2025) 601-603, https://doi.org/10.17776/csj.1746704.

ii. Christophe Rappold, et. al., Study of light hypernuclei in Europe: the hypertriton and nnΛ puzzles, EPJ Web of Conferences, 09007 (2023), https://doi.org/10.1051/epjconf/202329009007

iii. Takehiko R. Saito, et. al., Studies of three-and four-body hypernuclei with heavy-ion beams, nuclear emulsions and machine learning, Journal of Physics: Conference Series, **2586** 012148, (2023),https://iopscience.iop.org/article/10.1088/1742-6596/2586/1/012148

iv. Y. K. Tanaka et al., WASA-FRS EXPERIMENTS IN FAIR PHASE-0 AT GSI, Acta Physica Polonica B Proceedings Supplement, 16, 4-A27 (2023), https://doi.org/10.5506/APhysPolBSupp.16.4-A27

v. K. Nakazawa, *et al.,* Information and perspective for doubly-strange hypernuclei with nuclear emulsion detector, EPJ Web of Conferences **271**, 11005 (2022), https://doi.org/10.1051/epjconf/202227111005

vi. Takehiko R. Saito, for the WASA-FRS collaboration and the Emulsion-ML collaboration, Studies of hypernuclei with heavy-ion beams, nuclear emulsions and machine learning, EPJ Web of Conferences 271, 08003 (2022), https://doi.org/10.1051/epjconf/202227108003

vii. Manami Nakagawa, Ayumi Kasagi, Enqiang Liu, Hiroyuki Ekawa, Junya Yoshida, Wenbo Dou, Yan He, **Abdul Muneem**, Kazuma Nakazawa, Christophe Rappold, Nami Saito, Takehiko R. Saito, Shohei Sugimoto, Masato Taki, Yoshiki K. Tanaka, He Wang, Yiming Gao, Ayari Yanai, Masahiro Yoshimoto, Unique approach for precise determination of binding energies of hypernuclei with nuclear emulsion and machine learning, EPJ Web of Conferences 271, 11006 (2022), https://doi.org/10.1051/epjconf/202227111006

viii. Ayumi Kasagi, Enqiang Liu, Manami Nakagawa, Hiroyuki Ekawa, Junya Yoshida, Wenbo Dou, **Abdul Muneem**, Kazuma Nakazawa, Christophe Rappold, Nami Saito, Takehiko. R. Saito, Masato Taki, Yoshiki. K. Tanaka, He Wang, Precise measurement on the binding energy of hypertriton from the nuclear emulsion data using analysis with machine learning, Suplemento de la Revista Mexicana de Física, 30308122 (2022) 1–6,https://doi.org/10.31349/SuplRevMexFis.3.0308122

ix. Necla Cakmak, **Muneem Abdul**, First Forbidden β-Decay Logft Values of Neutron Rich Te Isotopes, ALKÜ Fen Bilimleri Dergisi, 2019, Special Issue (NSP 2018): page 45-51, https://dergipark.org.tr/en/pub/alku/issue/41490/492029

x. Necla Cakmak, Jameel-Un Nabi and **Muneem Abdul**, Beta Decay Half-lives of some Radioisotopes Produced in a Thermonuclear Reactor, CURRENT ACADEMIC STUDIES IN NATURAL SCIENCE AND MATHEMATICS SCIENCES-2018, Publisher: GeceKitaplığı, page 255-264.

Press releases regarding our manuscripts:

i. https://www.riken.jp/press/2025/20250124_2/index.html
ii. https://www.tohoku.ac.jp/japanese/2025/01/press20250127-01-Imaging.html
iii. https://www.uow.edu.pk/Views/Events/View?id=33091
iv. https://www.riken.jp/press/2025/20250829_1/index.html
v. https://www.gifu-u.ac.jp/news/research/uploads/20250829press02.pdf

Scholarly Profiles:

i. Google scholar: https://scholar.google.com/citations?user=90btlP0AAAAJ&hl=en
ii. ORCID: https://orcid.org/0000-0002-5316-7301
iii. ResearchGate: https://www.researchgate.net/profile/Abdul-Muneem-2?ev=hdr_xprf